

The Trace-Only Sobolev Obstruction for Non-Self-Adjoint Semiclassical Projections

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Status and source question

This packet gives a likely valid full affirmative answer to the conjecture in Figure 2 of Lafleche, arXiv:2302.04816v3 [1]. In the source, Theorem 1 proves maximal Sobolev nonregularity for self-adjoint projections. For non-self-adjoint idempotents it requires either $p \geq 2d/(d + s)$ or, in the remaining region, a uniform $\mathcal{L}^{2+\varepsilon}$ bound. The caption on page 7 says: “In this case, Theorem 1 needs more assumptions than a trace condition to be valid in the top-right zone. We conjecture however that the result holds there as well.”

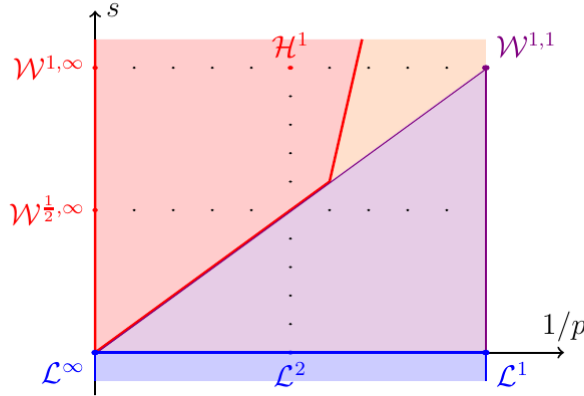


FIGURE 2. Regularity of non self-adjoint projection operators in Sobolev spaces $\mathcal{W}^{s,p}$ in dimension $d = 3$. In this case, Theorem 1 needs more assumptions than a trace condition to be valid in the top-right zone. We conjecture however that the result holds there as well.

Figure 1: Source page 7: Figure 2 and the complete conjectural caption.

Let $\mathsf{T}_z A = W_z A W_z^*$ denote the source’s quantum phase-space translation. The scaled Schatten norm is

$$\|A\|_{\mathcal{L}^p} = h^{d/p} \|A\|_{\mathfrak{S}_p}.$$

For $0 < s < 1$, the homogeneous quantum Sobolev seminorm is, up to the source’s positive normalization constant,

$$\|A\|_{\dot{\mathcal{W}}^{s,p}}^p = \gamma_{s,p} h^d \int_{\mathbb{R}^{2d}} \frac{\|\mathsf{T}_z A - A\|_{\mathfrak{S}_p}^p}{|z|^{2d+sp}} dz.$$

Proof Intuition

An oblique idempotent may have an arbitrarily large norm, so interpolation through an $\mathcal{L}^{2+\varepsilon}$ estimate loses the low- p region. That norm is unnecessary. Replace the idempotent P by the orthogonal projection Q onto the same range. If two ranges move under a unitary, the distance between their orthogonal projections is no larger, up to the sharp two-block factor $2^{1/p}$, than the distance between any idempotents having those ranges. Thus every translation difference of Q is controlled by the corresponding translation difference of P . The source's self-adjoint obstruction applies to Q and transfers back to P .

At order one the same mechanism appears infinitesimally. Relative to $\text{Ran } P \oplus (\text{Ran } P)^\perp$, the lower-left block of $[A, P]$ is precisely the lower-left block of $[A, Q]$. Self-adjointness of the generator supplies the opposite block of $[A, Q]$, again at a factor $2^{1/p}$.

The range-projection comparison

Lemma 1 (Schatten control of moving ranges). *Let P_0, P_1 be finite-rank idempotents on a Hilbert space, of the same rank, and let Q_i be the orthogonal projection onto $E_i = \text{Ran } P_i$. For $1 \leq p < \infty$,*

$$\|Q_1 - Q_0\|_{\mathfrak{S}_p} \leq 2^{1/p} \|P_1 - P_0\|_{\mathfrak{S}_p}.$$

For $p = \infty$, the same statement holds with constant 1.

Proof. Put $D = P_1 - P_0$. If $x \in E_0$, then $Dx = P_1x - x$, with $P_1x \in E_1$. Since Q_1x is the best approximation to x from E_1 ,

$$\|(I - Q_1)Q_0x\| \leq \|DQ_0x\|.$$

Consequently

$$Q_0(I - Q_1)Q_0 \leq Q_0D^*DQ_0$$

in positive-operator order, and the min-max principle gives

$$\|(I - Q_1)Q_0\|_{\mathfrak{S}_p} \leq \|DQ_0\|_{\mathfrak{S}_p} \leq \|D\|_{\mathfrak{S}_p}.$$

Interchanging the roles of the two ranges (and observing that $Dx = x - P_0x$ for $x \in E_1$) similarly gives

$$\|(I - Q_0)Q_1\|_{\mathfrak{S}_p} \leq \|D\|_{\mathfrak{S}_p}.$$

The square $(Q_1 - Q_0)^2$ commutes with Q_0 . Its restrictions to E_0 and $E_0^\perp$ are respectively

$$Q_0(I - Q_1)Q_0, \quad (I - Q_0)Q_1(I - Q_0).$$

It follows that

$$\|Q_1 - Q_0\|_{\mathfrak{S}_p}^p = \|(I - Q_1)Q_0\|_{\mathfrak{S}_p}^p + \|(I - Q_0)Q_1\|_{\mathfrak{S}_p}^p \leq 2\|D\|_{\mathfrak{S}_p}^p.$$

For the operator norm, take the maximum of the two diagonal blocks of the square instead of their trace sum. $\square$

Corollary 1 (Translation and commutator comparisons). *Let P be a finite-rank idempotent and Q the orthogonal projection onto $\text{Ran } P$. Then, for every unitary U ,*

$$\|UQU^* - Q\|_{\mathfrak{S}_p} \leq c_p \|UPU^* - P\|_{\mathfrak{S}_p}, \quad c_p = \begin{cases} 2^{1/p}, & p < \infty, \\ 1, & p = \infty. \end{cases}$$

If A is self-adjoint and the commutators belong to $\mathfrak{S}_p$, then

$$\|[A, Q]\|_{\mathfrak{S}_p} \leq c_p \|[A, P]\|_{\mathfrak{S}_p}.$$

Proof. The first claim is Lemma 1 applied to P and UPU^* , because the orthogonal projection onto $U \operatorname{Ran}(P)$ is UQU^* .

For the second claim, decompose the Hilbert space as $\operatorname{Ran} P \oplus (\operatorname{Ran} P)^\perp$. The idempotent and its range projection have block forms

$$P = \begin{pmatrix} I & B \\ 0 & 0 \end{pmatrix}, \quad Q = \begin{pmatrix} I & 0 \\ 0 & 0 \end{pmatrix}.$$

Writing $A_{21} = (I - Q)AQ$, compression gives

$$(I - Q)[A, P]Q = A_{21}, \quad [A, Q] = \begin{pmatrix} 0 & -A_{21}^* \\ A_{21} & 0 \end{pmatrix}.$$

Hence $\|[A, Q]\|_{\mathfrak{S}_p} = 2^{1/p} \|A_{21}\|_{\mathfrak{S}_p}$ for finite p (and equality without the factor for $p = \infty$), while $\|A_{21}\|_{\mathfrak{S}_p} \leq \|[A, P]\|_{\mathfrak{S}_p}$. $\square$

Solution of the Sobolev conjecture

Theorem 1. *Let $h \rightarrow 0$, and let P_h be trace-class idempotents satisfying*

$$P_h^2 = P_h, \quad h^d \operatorname{Tr}(P_h) = 1.$$

Let $1 < p \leq \infty$ and $0 < s \leq 1$, with $s \geq 1/p$. When $p = \infty$, assume $s > 0$. Then

$$\|P_h\|_{\dot{W}^{s,p}} \longrightarrow \infty.$$

Thus the extra uniform $\mathcal{L}^{2+\varepsilon}$ hypothesis in the non-self-adjoint part of Theorem 1 of the source is unnecessary throughout the Sobolev region conjectured in its Figure 2.

Proof. A trace-class idempotent has finite rank, and its trace equals its rank. Let Q_h be the orthogonal projection onto $\operatorname{Ran} P_h$. Then

$$Q_h^2 = Q_h = Q_h^*, \quad h^d \operatorname{Tr}(Q_h) = h^d \operatorname{rank}(P_h) = 1.$$

First take $0 < s < 1$. Apply Corollary 1 with the Weyl unitary implementing $\mathbb{T}_z$. After multiplying by the semiclassical scaling factor,

$$\|\mathbb{T}_z Q_h - Q_h\|_{\mathcal{L}^p} \leq c_p \|\mathbb{T}_z P_h - P_h\|_{\mathcal{L}^p} \quad \text{for every } z.$$

Integrating this pointwise inequality in the definition of the fractional seminorm yields

$$\|Q_h\|_{\dot{W}^{s,p}} \leq c_p \|P_h\|_{\dot{W}^{s,p}}.$$

The self-adjoint part of Theorem 1 in [1] says that the left side tends to infinity whenever $s \geq 1/p$ and $p > 1$. The right side must therefore tend to infinity as well.

It remains to take $s = 1$. The source's Corollary 2.1, applied to Q_h , gives for every $p > 1$

$$\frac{1}{h} \|[x, Q_h]\|_{\mathcal{L}^p} + \|[\nabla, Q_h]\|_{\mathcal{L}^p} \longrightarrow \infty.$$

Apply the commutator part of Corollary 1 componentwise to the self-adjoint generators x_j and $-i\partial_j$. The corresponding sum for Q_h is at most a dimension-dependent constant times the quantum $\dot{W}^{1,p}$ seminorm of P_h . Hence that seminorm diverges. If one of the commutators is not in $\mathfrak{S}_p$, the conclusion is immediate. $\square$

A Besov consequence below order one

The same idea also treats the source’s second-difference Besov norm whenever $0 < s < 1$. For completeness, write $\Delta_z = T_z - I$. On every Schatten class, the isometric group identity

$$\Delta_z = \frac{1}{2}\Delta_{2z} - \frac{1}{2}\Delta_z^2$$

iterates to

$$\Delta_z A = - \sum_{k=0}^{\infty} 2^{-k-1} \Delta_{2^k z}^2 A.$$

The remainder vanishes for each fixed Schatten operator. Minkowski’s inequality and the change of variables $w = 2^k z$ show that the first-difference Besov seminorm is at most

$$\frac{1}{2(1 - 2^{s-1})}$$

times the source’s second-difference seminorm. Conversely, $\|\Delta_z^2 A\|_{\mathfrak{S}_p} \leq 2\|\Delta_z A\|_{\mathfrak{S}_p}$. Combining these facts with Corollary 1 gives, for $0 < s < 1$ and $1 \leq p, q \leq \infty$,

$$\|Q_h\|_{\dot{\mathcal{B}}_{p,q}^s} \leq C_{s,p} \|P_h\|_{\dot{\mathcal{B}}_{p,q}^s}.$$

The self-adjoint Besov conclusion of Theorem 1 in the source therefore transfers to arbitrary idempotents for $0 < s < 1$: divergence holds when $s > 1/p$, or when $s = 1/p$ and $q < \infty$. This includes every critical case $s = 1/p$ with $p > 1$.

Verification and limitations

The proof is exact and finite-dimensional at its new step. The script `code/check_projection_inequality.py` is only a regression check: it tests the range-difference and commutator inequalities on random complex idempotents, including condition numbers up to 10^4 . It is not used as evidence in place of the proof.

On 2026-07-21, the run’s lightweight indexes and bounded web searches were checked using arXiv:2302.04816, the exact title and conjecture sentence, and combinations of “non-self-adjoint”, “idempotent”, “trace condition”, and “quantum Sobolev regularity”. The source itself and adjacent quantum Sobolev work were found, but no later exact solution. This does not guarantee novelty.

The theorem settles the Sobolev conjecture in Figure 2 and the Besov statement for $0 < s < 1$. It does not claim the separate second-difference Besov endpoint $s = 1, p = 1$, and it uses rather than reproves the source’s self-adjoint nonregularity theorem. A human reviewer should focus on Lemma 1 and on the compression/domain justification for the unbounded generators in the $s = 1$ argument.

References

- [1] L. Lafleche, *Optimal Semiclassical Regularity of Projection Operators and Strong Weyl Law*, arXiv:2302.04816v3 (2023), especially Theorem 1, Corollary 2.1, and Figure 2.